ST458 Assignment — Trading Strategy Report

Group F

1 Data Preparation and Feature Engineering

The dataset comprises 100 synthetic ETFs over 4 years (2010–2013) with daily OHLCV data and no missing values. The data is split chronologically: 2010–2011 for model training and 2012–2013 for out-of-sample (OOS) validation. No imputation is required. All features are computed from raw OHLCV columns using rolling windows, so the first $\sim$130 training days serve as a warm-up period.

Signal discovery. Cross-sectional Spearman IC analysis against forward returns at multiple horizons (1, 5, 10, 21, 42, 63 days) reveals a dual structure. The dominant signal is *medium-term mean reversion*: past 42–126 day returns predict 21-day forward returns negatively (IC ≈ -0.15 to -0.20, IC-IR > 1.0). Short-term (5–10d) momentum coexists weakly (IC $\approx +0.02$ to $+0.06$) at 1–5d horizons, but decays rapidly. PCA on the daily return matrix confirms one dominant common factor ($\sim$60% of variance explained, average pairwise $\rho \approx 0.61$), with a condition number of $\sim$95,000—directly mirroring the covariance instability problem from Lecture 6 and motivating both dollar-neutral construction and ℓ_1 regularisation.

Factor construction. 39 factors are engineered across four categories: (i) *Momentum/mean-reversion*: compound returns (7 windows, 5–126d), arithmetic mean returns (4 windows), Bollinger z-scores (21/42/63d), idiosyncratic momentum (30/42/60d, subtracting the cross-sectional mean to isolate asset-specific signals); (ii) *Volatility*: rolling std, skewness, kurtosis (5/10/20/60d), 60d high-low range; (iii) *Tail risk*: rolling min of multi-horizon returns (7/20/30/60d over 60d windows), positive-return skewness; (iv) *Volume*: lagged price-volume correlation (10/30d). Of these, 36 pass |IC-IR| ≥ 0.05 screening; the strongest are `mom_126d` (-1.65), `mom_30d` (-1.25), `mom_42d` (-1.22), `factor9` (-1.16).

2 Model Development

The strategy is developed as a two-layer pipeline: *Layer 1* combines the screened factors into a composite expected return $\hat{\mu}_i$ per ETF, and *Layer 2* translates $\hat{\mu}$ into a portfolio via constrained optimisation. Multiple alternatives for each layer are systematically compared.

Layer 1: Signal combination

IC-IR weighted rank composite (baseline). Factors are combined via cross-sectional ranks weighted by their individual IC-IR values, constructing a top/bottom-decile long-short portfolio. With 21-day rebalancing at 5× scale, this achieves OOS log wealth of 1.56 (SR = 3.88, MDD = 9.4%). This treats factors independently: the weight for `mom_42d` ignores its $\rho > 0.8$ correlation with `mom_30d`.

Ridge regression (selected model). Ridge estimates factor weights *jointly*, handling inter-factor correlation ($\rho \approx 0.5$–0.9 among momentum variants) via ℓ_2 regularisation ($\alpha = 10$, selected by time-series CV with 21-day gap; CV IC = +0.18). The fitted coefficients reveal structure invisible to independent weighting: `mom_20d` (-0.064) and `mom_20d_mean` ($+0.062$) receive opposing signs, effectively trading on return convexity (the difference between compound and arithmetic mean returns). The model is fitted once on all training data and frozen for the test period.

Gradient boosted trees. LightGBM, XGBoost, and Random Forest are trained on the same 36 features with hyperparameters tuned via randomised grid search over walk-forward CV folds (Lecture 8 pipeline). Two ensemble methods—simple averaging and cross-sectional rank averaging of predictions—are also tested. OOS ICs range from +0.222 (XGBoost) to +0.249 (average ensemble), with LightGBM at +0.246.

Neural networks. Four architectures are tested: a linear MLP ($\equiv$ OLS), a nonlinear MLP (2 hidden layers, ReLU, dropout; Lecture 9), a 2-layer LSTM on 21-day sequences (Lecture 10), and a two-headed LSTM concatenating raw OHLCV hidden states with engineered features. The linear MLP (IC = +0.223) matches Ridge; deeper models show diminishing returns (LSTM IC = +0.08), consistent with the low SNR of financial data (Lecture 7).

Layer 2: Portfolio construction

Following Lecture 6, the gross-exposure constrained mean-variance problem is solved:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \left(-a\,\mathbf{w}^\top \hat{\boldsymbol{\mu}} + \tfrac{a^2}{2}\,\mathbf{w}^\top \hat{\Sigma}\,\mathbf{w} \right) \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 0, \;\; \|\mathbf{w}\|_1 \leq c, \;\; |w_i| \leq 0.10,$$

where $\hat{\Sigma}$ is estimated from a rolling 126-day return window with $10^{-6} I$ regularisation. The dollar-neutrality constraint hedges the dominant common factor from PCA.

The ℓ_1 constraint stabilises the portfolio: by Proposition 2 of Lecture 6, excess risk from using $\hat{\Sigma}$ instead of Σ is bounded by $2c^2\|\hat{\Sigma} - \Sigma\|_\infty$, which remains well-controlled since each entry of $\hat{\Sigma}$ is a low-dimensional estimation problem even when the spectral properties of $\hat{\Sigma}$ are poor. The per-asset cap $|w_i| \leq 0.10$ prevents concentration. A rank-based equal-weight long-short portfolio (top/bottom decile) serves as a baseline avoiding covariance estimation.

Fair comparison at unit scale

To isolate signal quality from leverage effects, all models are evaluated at scale $= 1$ (no leverage amplification) through both portfolio construction methods. Figure 1 shows the three core signal-portfolio combinations under identical conditions. The improvements are additive: switching from IC-IR weighting to Ridge improves log wealth from 0.559 to 0.586 (better signal combination), and switching from rank-based to Markowitz portfolio construction further lifts it to 0.801 (better use of predicted return magnitudes and covariance structure). The Sharpe ratio also improves from 4.53 to 4.95, while maximum drawdown remains below 3.5% at unit scale.

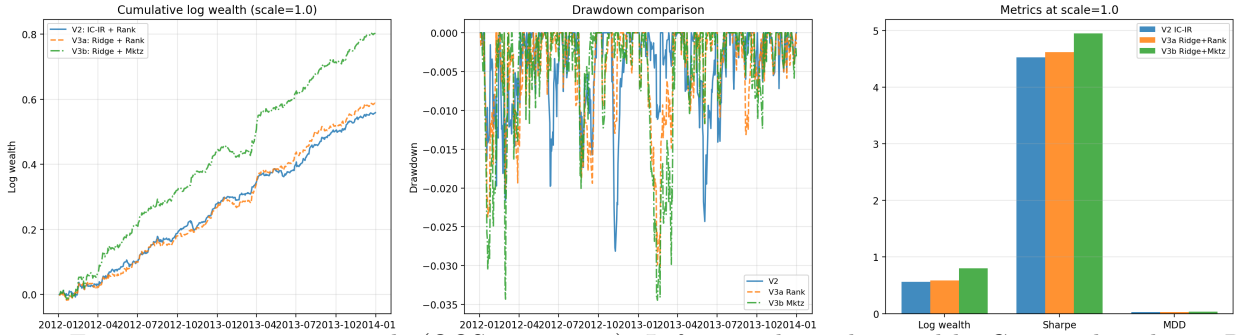

Figure 1: Fair comparison at unit scale (OOS 2012–2013). Left: cumulative log wealth. Centre: drawdown. Right: summary metrics. Both improvements (Ridge signal and Markowitz construction) contribute additively.

Full model comparison across 10 signal alternatives

All 10 signal models are fed through the same Markowitz optimiser ($a = 0.5$, $c = 1.0$) for a controlled comparison (Figure 2). A key finding: *higher IC does not guarantee higher portfolio performance*. The average ensemble achieves the highest IC (+0.249) but only 4th-highest Markowitz log wealth (0.691). Ridge (IC = +0.226) produces the best Markowitz log wealth (0.794) because its smoother, more continuous $\hat{\mu}$ vector is better conditioned for the mean-variance optimiser—tree-based and ensemble predictions contain more cross-sectional noise that $\hat{\Sigma}^{-1}\hat{\mu}$ amplifies. This is visible in panel (e): Ridge sits well above the trend line relating IC to portfolio performance. Panel (f) confirms the Markowitz uplift (Δ log wealth when switching from rank-based to Markowitz construction) is largest for Ridge (+0.20) and near-zero for Random Forest and LightGBM, whose noisier predictions receive no benefit from the optimiser.

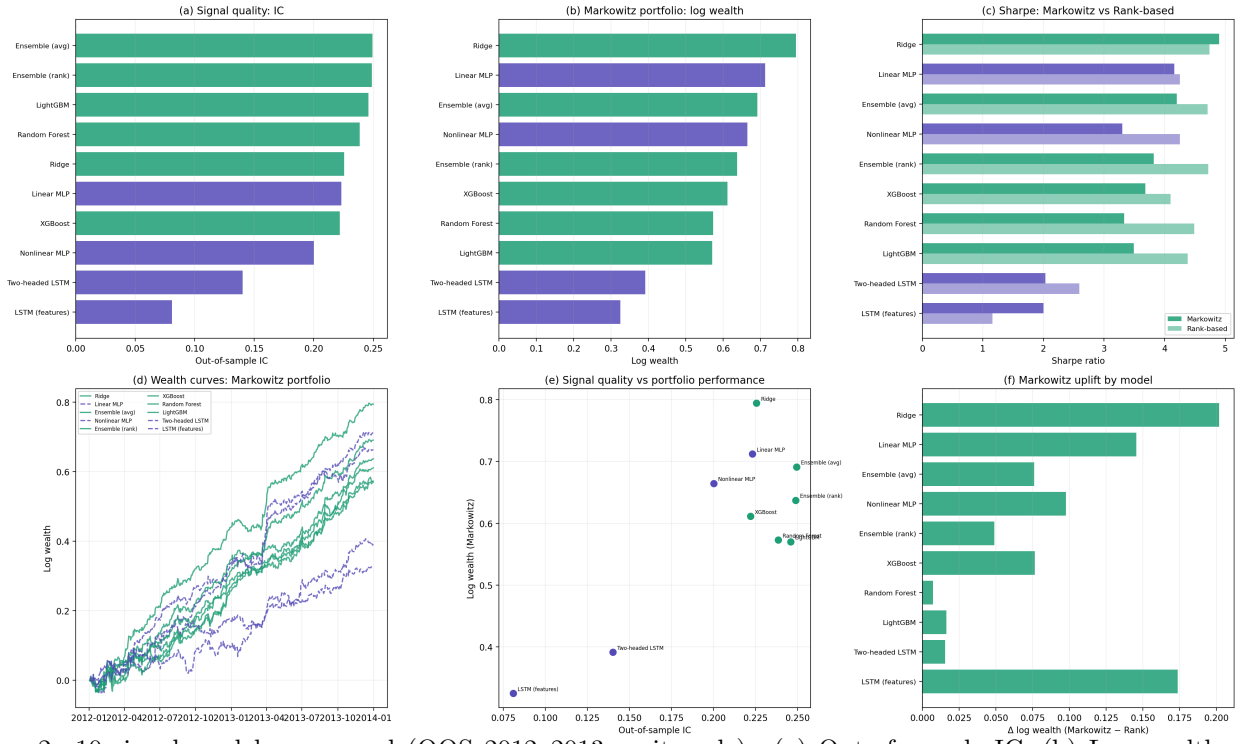

Figure 2: 10 signal models compared (OOS 2012–2013, unit scale). (a) Out-of-sample IC. (b) Log wealth under Markowitz construction. (c) Sharpe ratio: Markowitz vs rank-based. (d) Wealth curves. (e) IC vs portfolio log wealth—Ridge outperforms despite lower IC. (f) Markowitz uplift by model.

Pairs trading experiment

Following Lectures 3–4, cointegration-based pairs trading is also explored as an alternative strategy. The top 20 pairs by return correlation are identified and the Engle–Granger ADF test is applied to their log-price spreads; four pairs pass at the 5% critical value (SYDR–WQX, JBDX–JZKP, CDM–HJC, KRV–WYLR). ADF pre-tests confirm all 8 constituent symbols are plausibly I(1), validating the cointegration testing assumptions. A Bollinger-band strategy with rolling z-score entry/exit thresholds is implemented on the stationary spreads. However, this approach is not adopted for the final submission: with only 4 tradeable pairs the strategy has very low breadth (the Fundamental Law of Active Management from Lecture 5 implies IR $\approx$ IC $\times\sqrt{\text{breadth}}$, and breadth ≤ 4 severely limits achievable IR), gross exposure is capped at 0.25 for safety, and the cointegration relationships estimated from just 2 years of training data carry substantial risk of structural break out-of-sample.

3 Final Strategy and Rationale

Architecture. The submitted strategy (`GroupF.py`) implements Ridge as the signal layer and constrained Markowitz as the portfolio layer. On each rebalance day (every 21 trading days): (1) compute 36 features from the rolling price buffer, (2) apply the frozen Ridge model to obtain $\hat{\mu}$, (3) estimate $\hat{\Sigma}$ from 126 days of returns, (4) solve the constrained optimisation via CVXPY (scipy SLSQP fallback), and (5) trade toward the target. Drawdown protection halves exposure beyond 5% drawdown and quarters it beyond 10%.

Leverage configuration. Two parameters control leverage: a *scale* multiplier applied post-optimisation and the ℓ_1 bound c. Since `walk_forward.py` tracks positions in fixed dollar amounts, wealth-relative leverage *decreases* as wealth accumulates: a \$6 gross position is 6× leverage at wealth = \$1 but only 1× at wealth = \$6. Figure 3 reports a grid search over 12 (scale, c) combinations; the theoretical ceiling scale $\times$ c is rarely reached in practice.

Leverage configuration grid: out-of-sample performance (2012–2013, 502 trading days)

scale	c	s×c	logW	SR	MDD	Ann Vol	avg GE($)	max GE($)	avg GE/W	max GE/W	min W	final W
5	3.0	15	2.373	4.74	22.3%	1.031	14.07	18.75	4.61	17.65	0.80	10.73
5	2.0	10	2.171	4.87	22.4%	0.801	9.26	12.54	3.44	11.60	0.82	8.77
5	1.5	8	2.062	4.83	16.4%	0.713	7.22	9.36	2.54	8.10	0.91	7.87
3	3.0	9	1.966	4.88	13.6%	0.632	8.83	11.25	3.43	9.69	0.88	7.14
3	**2.0**	**6**	**1.743**	**4.84**	**13.8%**	**0.489**	**5.76**	**7.53**	**2.65**	**6.45**	**0.89**	**5.72**
3	1.5	4	1.656	4.90	11.2%	0.434	4.43	5.62	1.93	4.94	0.95	5.24
2	3.0	6	1.645	4.91	9.1%	0.428	6.02	7.50	2.73	6.71	0.92	5.18
2	2.0	4	1.439	4.88	9.3%	0.330	3.92	5.02	2.07	4.18	0.93	4.21
2	1.5	3	1.343	4.84	8.0%	0.294	3.02	3.74	1.58	3.32	0.96	3.83
1	3.0	3	1.128	4.91	4.7%	0.214	3.01	3.75	1.77	3.36	0.96	3.09
1	2.0	2	0.971	4.89	4.7%	0.169	2.01	2.51	1.31	2.24	0.96	2.64
1	1.0	1	0.790	4.99	3.4%	0.121	1.01	1.25	0.69	1.13	0.99	2.20

Figure 3: Leverage configuration grid (OOS 2012–2013, 502 days). "avg GE/W" and "max GE/W" report realised leverage relative to current wealth. The highlighted row (scale = 3, $c = 2$) is the selected submission configuration.

Selected configuration and rationale. The configuration (scale $= 3$, $c = 2$, $a = 0.5$, hold $=$ 21d) is selected, yielding OOS log wealth of 1.74, Sharpe ratio of 4.84, max drawdown of 13.8%, and minimum wealth of 0.89. The 21-day rebalance period is chosen to match the dominant signal horizon (42–126d lookback predicting 21d forward returns), while limiting turnover at 5 bps transaction cost. More aggressive settings (e.g. scale $= 5$, $c = 3$: logW = 2.37, MDD = 22%, min wealth = 0.80) achieve higher expected returns but carry greater tail risk across the 100 test realisations. The scoring rule averages log wealth with a -10 floor penalty for zero wealth, making the cost of a single ruin event far larger than the benefit of higher average returns. The moderate configuration maintains average realised leverage of $\sim$2.65× (well below the 6× theoretical ceiling), balancing expected growth against this asymmetric penalty structure. The risk aversion parameter $a = 0.5$ is chosen because it is insensitive in practice: the grid search over $a \in \{0.5, 1.0, 2.0\}$ shows nearly identical results at each c, since the per-asset cap and ℓ_1 constraint dominate the shape of the feasible set.